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PROJECT AND TEAM INFORMATION

Project Title

Logistics And Delivery Management

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PHASE-II PROJECT PROGRESS DESCRIPTION

Project Abstract

Logistics and Delivery Management System is a desktop application that helps users perform logistics and delivery management operations in an efficient manner. This system acts as a medium for managing customer management, deliveries, delivery people management, vehicles, shipments, and delivery status. The main purpose of the system is to minimize manual effort and organize the whole process of tracking and managing the deliveries. The users will be able to update shipments, assign deliveries, manage delivery people and vehicles, and track the current status of shipments using an interactive graphical user interface. The development of the application involves C++ and Qt and is implemented from an object-oriented perspective. Various modules are represented in the form of classes in order to increase the flexibility and reusability of the code. Data structures and relevant validation methods are used in the system. The final system is expected to provide a user-friendly interface that would allow users to get necessary logistics information and perform delivery management operations.

Updated Project Approach and Architecture

- The Logistics and Delivery Management System is being developed as a desktop-based application using C++ and the Qt framework. The project follows a modular, object-oriented architecture that separates the graphical user interface, application logic, and data management. The Qt interface includes a dashboard, forms, tables, and navigation options for managing customers, shipments, delivery personnel, vehicles, and delivery status. C++ classes are designed to represent these entities and handle their respective operations, making the application organized, reusable, and easier to maintain.
- The application uses Qt widgets to create an interactive and user-friendly interface. Qt's signals and slots mechanism connects user actions, such as button clicks and form submissions, to the corresponding functions. Object-oriented concepts, including encapsulation, abstraction, inheritance, and polymorphism, are applied wherever appropriate to improve code structure and reusability. The application also includes input validation and error handling to prevent incorrect data entry. Development is carried out module by module, followed by integration and functional testing to verify that customer management, shipment handling, vehicle assignment, and delivery tracking work together correctly. The main focus is to develop a reliable, efficient, and easy-to-use system that simplifies logistics operations and improves delivery management.

Tasks Completed

Task Completed	Team Member
Completed the user, customer, and authentication modules, along with searching, sorting, and binary search algorithms.	Kush Agarwal
Completed the delivery, driver, and logistics modules, along with queue and priority queue implementation.	Bhupendra Singh Thapa
Completed file handling and validation modules, along with linked list and stack implementation.	Raghav Varshney
Developed the Qt-based GUI with a main window, dashboard, widgets, and navigation features.	Ishan Jakhmola

Challenges/Roadblocks

Development has presented a number of major problems, not least in the form of the complexity inherent to the different backend modules. This covers everything from data structures and authentication to the management of customers and deliveries. The plan is to put an end to this by breaking the project down into its constituent modules and assigning tasks on an individual basis.

Then there is the matter of the Qt-based graphical user interface and how it is to be integrated with the backend. It is no easy task to ensure that what is done in the UI is relayed correctly to the backend classes; we will address this by way of Qt signals and slots to link the UI components to the right functions.

Consistency of information is also a concern, especially with logistics, so file handling will see some improvement in the code along with the addition of sound error handling and input validation.

As for the team, one can expect integration issues to arise when combining the code they have written. To prevent this, there will be regular communication and a good deal of testing to go along with the distribution of work. In the same vein, both integration and module-level testing will be ongoing.

Tasks Pending

Pending Task	Assigned To
Feature Development	Kush Agarwal, Bhupendra Singh Thapa
Module Integration	Ishan Jakhmola, Bhupendra Singh Thapa
Input Validation	Raghav Varshney
Error Handling	Raghav Varshney, Kush Agarwal
Testing and Debugging	All Team Members
Project Documentation	All Team Members

Project Outcome/Deliverables

At present the project is in the development and integration phase, with a basic architecture in place. To make for easier collaboration and a smoother process, the backend modules have been decoupled.

Some of these are already complete, namely the AuthManager and Customer modules. The rest of the backend consists of the User, Delivery, Driver, LogisticsSystem, FileManager and Validator modules, not to mention the data structures and algorithms required to handle information processing.

On the front end, a graphical interface has been put together with the Qt framework; it is being built on its own and includes the window, dashboard, widgets and means of navigation. What remains is to see the development of the connecting modules through to completion so as to link the GUI with the backend, then test the principal functionality and put right any errors that turn up.

Progress Overview

- The project is currently at the development and integration stage.
- The fundamental components of the project have been developed, and the backend has been divided into modules to simplify development and interaction.
- The AuthManager and Customer modules have already been implemented.
- The remaining backend modules include User, Delivery, Driver, LogisticsSystem, FileManager, and Validator, along with the algorithms and data structures required for processing and storing information.
- The graphical user interface (GUI), developed using the Qt framework, includes a window, dashboard, widgets, and navigation tools. It is being developed separately from the backend.
- The remaining tasks involve completing the modules that connect the GUI with the backend, testing the main features of the application, and fixing the identified bugs.

Codebase Information

Category	Description
Repository Link	https://github.com/agarwkush/PBL
Codebase Overview	The GitHub repository contains the project's source code and related files. It helps the team organize code, track changes, collaborate on development, and maintain project updates.

Testing and Validation Status

Module Tested	Result	Remarks
User Authentication	Pass	Valid credentials verified.
Customer Data Management	Pass	Customer records managed successfully.
Driver Allocation and Delivery Tracking	Pass	Driver assignment and status updates verified.
File Handling	Pass	File read and write operations verified.
Data Structure Implementation	In Progress	Testing data structure operations.
Searching and Sorting Algorithms	In Progress	Verifying search and sorting accuracy.
Input Validation and Error Handling	Pass	Invalid inputs handled correctly.
Qt GUI Integration	In Progress	Testing GUI and backend integration.

Deliverables Progress

-Project Planning: Identified the key requirements for the Logistics and Delivery Management System.

- UI Design: Developed the first draft of the app interface with Qt.

Module Design: Designed the core modules for delivery, shipping information, and logistics.

- System Architecture: Defined the basic architecture of the application and how the different parts interact.

- Implementation Progress: Initiated structuring of the source code and development of the application's initial components.

- Project Status: The project is in development status and further implementation is ongoing.

- Work to be done: Development of core functionality, integration of modules, testing, error fixing and documentation preparation.